LLC-type series resonant convertor with PWM control

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Indexing terms: Computer simulation, Control characteristics, Modes of operation, Series resonant convertor, Steady-state analysis

Abstract: The steady-state analysis and performance characteristics of the *LCC*-type series resonant convertor (SRC) are presented. The analysis is based on the PWM control scheme for both continuous and discontinuous conduction mode of operations. By using the same control techniques, it is shown that the control characteristics of the *LLC*-type SRC are substantially improved over those of the conventional SRC. Convertor component stress curves are reported, and compared, under various load ranges, with the conventional SRC. Also computer simulation and experimental data are given to verify the theoretical results.

1 Introduction

The ever increasing demand for smaller size and lighter weight power convertors, and interest in high frequency DC–DC resonant convertors has been steadily rising since the early 1980s [1–14]. Because of the reduced switching losses due to the sinusoidal behaviour of resonant convertors, it is possible to operate such convertors at high frequency to reduce the size of their reactive components. Many of today's resonant topologies use high-order resonant tanks in order to improve the control characteristic curves, and utilise the parasitic components of the transformer leakage inductance and transistor junction capacitors [6–8]. Among these are the series–parallel *LCC*-type and *LLC*-type resonant convertors. Detailed analysis of the *LLC*-type series resonant convertor (SRC-*LLC*), which is the subject of this paper, has been reported in Reference 10.

There are many output regulation control schemes available in the design of DC–DC resonant convertors. Such control techniques may be broadly classified as variable-frequency [1, 2] and constant-frequency controls [3, 4]. Even though variable-frequency control can be easily implemented in practice, the EMI noise generated in such a convertor may be wideband if a large range of switching frequency is required for output regulation. Other disadvantages associated with the large range of switching frequency are poor crossregulation in multiple-output convertors, and higher component losses at the upper switching frequencies.

The problems mentioned above can be minimised by using the constant-frequency control technique in the convertor design. The constant-frequency control technique can be implemented in two ways: PWM control and phase-shift control. In the PWM control scheme, a minimum of two transistor switches are needed, such as those in the half-bridge configuration, and each transistor switch is operated at less than 50% duty ratio. With phase-shift control, the method can be implemented using two pairs of transistor switches, with each pair of switches being turned *off* and *on* at 50% duty ratio. These two pairs of switches are arranged in such a way that the phase-shift between them is used for convertor control, such as those in the full-bridge configuration.

It has been shown that, in the case of PWM control, the convertor voltage gain of the conventional series resonant convertor is load independent only when the convertor is operated in the continuous conduction mode [3]. As the convertor output changes from full load to no load, its operation may change from the continuous to the discontinuous conduction mode. Consequently, the range of duty ratio needed for output regulation may be prohibitively large. The proposed third-order convertor, known as the *LLC*-type SRC, reduces the range of required duty ratio for output regulation.

In this paper, we present an analysis and design of a PWM-controlled *LLC*-type SRC operating in continuous and discontinuous conduction modes. By using both time domain and state-plane analyses, the steady-state expressions can be derived. Based on these analyses, a set of control characteristic curves for the *LLC*-type SRC is reported. It has been shown that a conventional second-order SRC [1] and the *LLC*-SRC [2], when operating in the continuous conduction mode, yield maximum gain of unity when their transistors are switched at the resonant frequencies. In order to obtain the maximum range of output voltage gain (between zero and one), the switching frequency is chosen to be equal to the resonant frequency when operating in the continuous conduction mode. We give the performance characteristics of the convertor operating in this condition and compare them with those of the conventional SRC using PWM control.

2 Basic circuit operation

An idealised half-bridge *LLC*-type SRC is shown in Fig. 1. For PWM control, transistors T_1 and T_2 are switched alternately at less than 50% duty ratio. The simplified equivalent circuit is shown in Fig. 2, where all diodes and transistors are assumed ideal. In this circuit, when the switch is in T_1 or T_2 position, diode D_2 or D_1 is open,

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Paper 9900G (P6), received 24th May 1993

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respectively. When it is in the open position, if either D_1 or D_2 is on, the convertor is operated in the continuous conduction mode. Otherwise, it is in the discontinuous conduction mode.

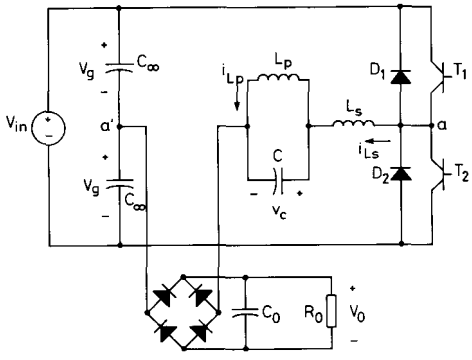

Fig. 1 *Half-bridge circuit of the LLC-type series resonant convertor*

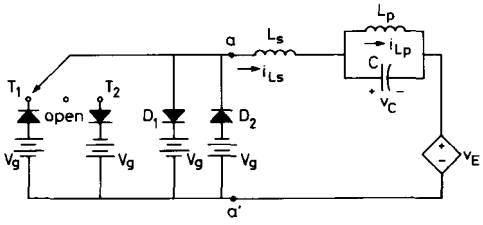

Fig. 2 *Convertor equivalent circuit*

In the continuous conduction mode of operation, the resonant tank circuit of the convertor is driven by an equivalent two-state source $(+V_g, -V_g)$ and is terminated by a two-state voltage sink $v_E(+V_0, -V_0)$, as shown in Fig. 2. Over a switching period, there exist four possible topological circuit modes for the convertor, as indicated in its steady-state trajectory given in Fig. 3 and its corresponding topological circuits shown in Fig. 5, where Z_0 is the characteristic impedance defined as $\sqrt{(L/C)}$. The detailed derivation of Fig. 3 will be presented in the next Section. Since the trajectory of Fig. 3 possesses odd symmetry about the origin, where $i_L = i_{Ls} - i_{Lp}$, only the response over a half switching period is needed for the steady-state analysis. The basic circuit operation can be described as follows: the cycle starts at t_0 when T_1 is turned on and the convertor is in circuit mode $M1$. The circuit mode changes from $M1$ to $M2$ at t_1 when transistor T1 is turned off and i_{Ls} is forced to commutate from T_1 to D_2. In the time interval $[t_1, t_0 + T_s/2]$, both T_1 and T_2 are turned off, hence, the switch is in the open-circuit position. In circuit mode $M2$, the amplitude of i_{Ls} decreases and reaches zero at t_2. For the continuous conduction mode of operation, the capacitor voltage, v_C, at $t = t_2$ is greater than $V_g + V_0$, causing i_{Ls} to reverse its direction through diode D_1. Consequently, diode D_1 is naturally turned on at t_2 and the convertor is in circuit mode $M4$. Diode D_1 stays on until transistor T_2 is turned on at $t = t_0 + T_s/2$. At this time, the switching operation for the half switching period is completed and the second half switching period starts at $t = t_0 + T_s/2$ and can be similarly described.

In the discontinuous conduction mode of operation, the equivalent voltage source at terminals a–a' has three states; $+V_g$, $-V_g$ and open circuit. In this operating mode, the convertor possesses two resonant frequencies: ω_0 when the driving source equals $+V_g$ or $-V_g$, and ω_p when the source is open, where $\omega_0 = 1/\sqrt{(LC)}$, $\omega_p = 1/\sqrt{(L_p C)}$, and $L = L_p L_s/(L_p + L_s)$. The state-plane diagram for the discontinuous conduction mode of operation is shown in Fig. 4.

The convertor operations for the continuous and discontinuous conduction modes are the same in the time interval $[t_0, t_2]$. In the case of discontinuous conduction mode, at $t = t_2$, the capacitor voltage v_C is less than $V_g + V_0$ when i_{Ls} equals zero. Thus, both D_1 and D_2 are turned off and terminals a–a' are open-circuited. The residual energy stored in the parallel tank circuit C–L_p oscillates at ω_p and the convertor is in circuit mode $M5$. At t_3, T_2 is turned on and i_{Ls} starts increasing in the opposite direction. At this time the upper half of the switching cycle is completed. The circuit modes $M1$, $M2$, $M3$, $M4$, and $M5$ are shown in Fig. 5.

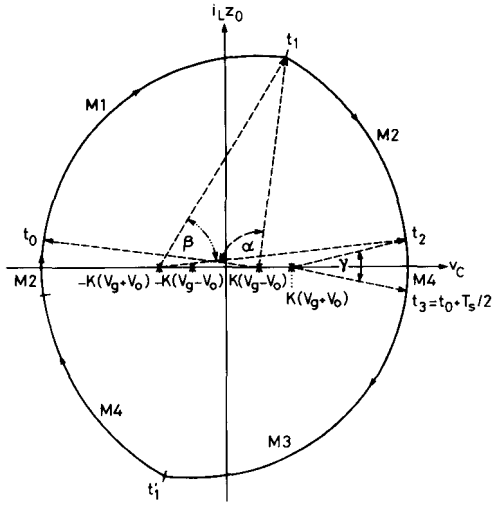

Fig. 3 *Steady-state trajectory in the continuous conduction mode*

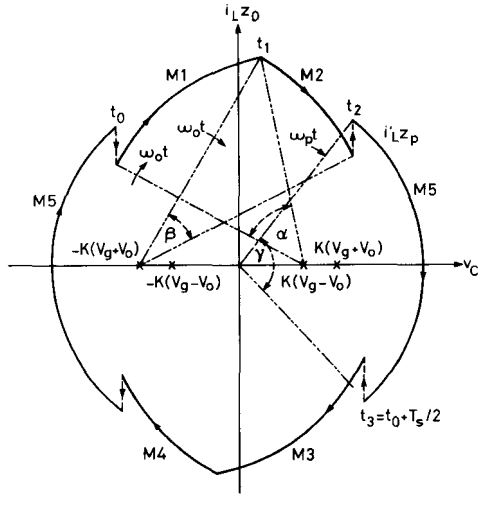

Fig. 4 *Steady-state trajectory in the discontinuous conduction mode*

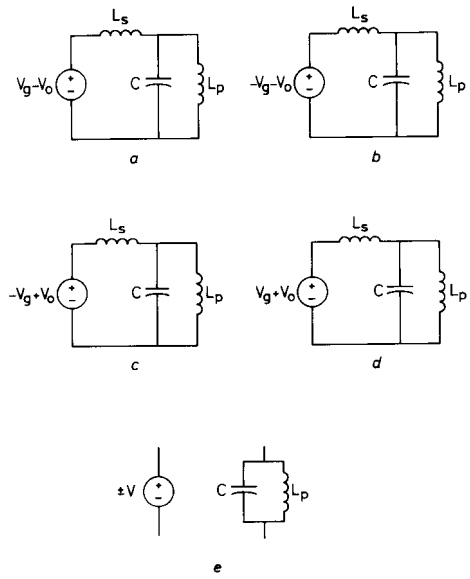

Fig. 5 *Five topological circuit modes*
a $M1$: T_1 on
b $M2$: D_2 on
c $M3$: T_2 on
d $M4$: D_1 on
e $M5$: All switching devices off

3 Steady-state response

3.1 Continuous conduction mode of operation

When the convertor operates in the continuous conduction mode (CCM), the voltage source has one of the two states; $+V_g$ or $-V_g$, depending on whether transistor T_1 or T_2 is turned on, respectively. When both transistors are off, the state $+V_g$ or $-V_g$ occurs, depending on whether the direction of i_{Ls} is negative or positive. The current i_{Ls} continuously flows in one direction through diode D_1 (or D_2), reaches zero, and continues to flow in the opposite direction through diode D_2 (or D_1). During the time when both transistors are turned off, current i_{Ls} does not remain zero because the amplitude of capacitor voltage v_C is greater than $V_g + V_0$ at the instant when $i_{Ls} = 0$. Natural commutation takes place between diodes D_1 and D_2 at this time.

As shown in Fig. 3, the mode sequence for the continuous conduction mode of operation is $M1$–$M2$–$M4$–$M3$–$M4$–$M2$. The transition from $M2$ to $M4$ or $M4$ to $M2$ takes place because natural commutation between diodes D_1 and D_2 occurs. In this mode of operation, the convertor has only one resonant frequency, ω_0. In this paper, our analysis is limited to the case in which the switching frequency is equal to the natural resonant frequency, ω_0, of the convertor.

The state equations for circuit modes $M1$, $M2$, $M3$ and $M4$ are given by

$$\frac{di_{Ls}}{dt} = \frac{V_s - v_C}{L_s} \tag{1}$$

$$\frac{di_{Lp}}{dt} = \frac{v_C}{L_p} \tag{2}$$

$$\frac{dv_C}{dt} = \frac{i_{Ls} - i_{Lp}}{C} = \frac{i_L}{C} \tag{3}$$

where V_s equals $(V_g - V_0)$, $(-V_g - V_0)$, $(-V_g + V_0)$, $(V_g + V_0)$, for modes M_1, $M2$, $M3$, and $M4$, respectively. From eqns. 1–3, the state-plane equation for circuit modes $M1$, $M2$, $M3$ and $M4$ can be written as

$$\frac{di_{nL}}{dv_{nC}} = \frac{kV_{ns} - v_{nC}}{i_{nL}} \tag{4}$$

where

$$k = \frac{L_p}{L_s + L_p} \quad i_{nL} = \frac{i_L Z_0}{V_g} \quad i_L = i_{Ls} - i_{Lp}$$

$$v_{nC} = \frac{v_C}{V_g} \quad M = \frac{V_0}{V_g} \quad Z_0 = \sqrt{\left[\frac{L_p L_s}{(L_p + L_s)C}\right]}$$

and V_{ns} is the normalised voltage which equals $1 - M$, $-1 - M$, $-1 + M$, and $1 + M$ for mode $M1$, $M2$, $M3$ and $M4$, respectively. From eqn. 4, the trajectory in each mode of operation can be written as

$$i_{nL}^2 + [v_{nC}(t) - kV_{ns}(t)]^2 = R_i^2(t_i) \tag{5}$$

where

$$R_i^2(t_i) = I_{nL}^2(t_i) + [V_{nC}(t_i) - kV_{ns}(t_i)]^2$$

For $i = 0, 1, 2$ and 3.

In order to determine the transition from $M2$ to $M4$ (or from $M4$ to $M2$), we need an expression for $i_{nLs}(i_{Ls} Z_0/V_g)$ which can be derived from the convertor circuit and is given by

$$\begin{aligned} i_{nLs}(t) = {} & I_{nLs}(t_i)(1 - k + k\cos\omega_0 t') + \frac{Z_0}{L_p + L_s} V_{ns} t' \\ & + \frac{kZ_0}{\omega_0 L_s} V_{ns} \sin\omega_0 t' + \frac{Z_0}{\omega_0 L_s} V_{nC}(t_i) \sin\omega_0 t' \\ & + k[I_{nLs}(t_i) - I_{nL}(t_i)](1 - \cos\omega_0 t') \end{aligned} \tag{6}$$

where

$$\omega_0 = \sqrt{\left(\frac{L_p + L_s}{L_p L_s C}\right)} \quad \text{and} \quad t' = t - t_i$$

and t_i is the instant at which the converter enter into circuit modes $M1$, $M2$, $M3$ or $M4$. From the state–plane trajectory of the continuous conduction mode shown in Fig. 3, at $t = t_2$, $i_{nLs}(t_2) = 0$. Thus we have

$$\begin{aligned} i_{nLs}(t_2) = {} & I_{nLs}(t_1)(1 - k + k\cos\beta) + \frac{Z_0}{L_p + L_s} V_{ns} \frac{\beta}{\omega_0} \\ & + \frac{kZ_0}{\omega_0 L_s} V_{ns} \sin\beta + \frac{Z_0}{\omega_0 L_s} V_{nC}(t_1) \sin\beta \\ & + k[I_{nLs}(t_1) - I_{nL}(t_1)](1 - \cos\beta) \end{aligned} \tag{7}$$

Eqn. 7 can be numerically solved to find β which is the conduction angle of diode D_2. The capacitor voltage v_{nC} can also be calculated at $t = t_2$ to determine whether the next circuit mode should be $M4$ for the continuous mode or $M5$ for the discontinuous mode. From the trajectory of the continuous conduction mode given in Fig. 3, we have

$$\alpha + \beta + \gamma = \pi \tag{8}$$

For steady-state operation, the following cyclic conditions must be satisfied

$$i_{nLs}(t_0) = -i_{nLs}(t_0 + T_s/2) \tag{9}$$

$$v_{nC}(t_0) = -v_{nC}(t_0 + T_s/2) \tag{10}$$

$$i_{nL}(t_0) = -i_{nL}(t_0 + T_s/2) \tag{11}$$

To obtain the steady-state solutions, we need additional information on the load condition which will be given in Section 4.

3.2 Discontinuous conduction mode of operation

In this mode of operation, the resonant circuit is driven by the voltage source which has one of the three states; $+V_g$, open, or $-V_g$. The state V_g or $-V_g$ is controlled by duty ratio d, depending on whether transistor T_1 or T_2 is turned on, respectively. When both transistors are turned off, the state $+V_g$ or $-V_g$ occurs, depending on whether i_{Ls} is negative or positive. The open-circuit state exists when i_{Ls} reaches zero. In this case, both diodes D_1 and D_2 are turned off, since the magnitude of capacitor voltage v_C is less than $V_g + V_0$. The current I_{Ls} remains zero in the open-circuit state until transistor T_1 or T_2 is turned on.

As shown in Fig. 4, the mode sequence for the discontinuous conduction mode of operation is $M1$–$M2$–$M5$–$M3$–$M4$–$M5$. The transition from $M2$ to $M5$ occurs when v_C is less than $V_g + V_0$ at the instant when i_{Ls} equals zero. There are two resonant frequencies associated with the convertor circuit: ω_0 in modes $M1$, $M2$, $M3$ and $M4$, which is also the natural resonant frequency of the convertor, and ω_p in mode $M5$. As can be seen from Fig. 4, the state-plane trajectory in the first half of the switching period possesses discontinuity jumps at t_0, t_2 and t_3 due to different scaling factors used in the state-plane diagram.

The state equations for circuit modes $M1$, $M2$, $M3$ and $M4$ are the same as eqns. 1–4. The state equations for $M5$ are given by

$$i_{Ls} = 0 \tag{12}$$

$$\frac{di_{Lp}}{dt} = \frac{v_C}{L_p} \tag{13}$$

$$\frac{dv_C}{dt} = \frac{-i_{Lp}}{C} \tag{14}$$

From eqns. 12–14, the state-plane equation for circuit mode $M5$ can be written as

$$\frac{di'_{nL}}{dv_{nC}} = \frac{v_{nC}}{-i'_{nL}} \tag{15}$$

where

$$i'_{nL} = -i_{Lp}\frac{Z_p}{V_g} \quad Z_p = \sqrt{\left(\frac{L_p}{C}\right)}$$

From eqn. 15, the state-plane trajectory for circuit mode $M5$ can be written as

$$i'^2_{nL}(t - t_2) + v^2_{nC}(t - t_2) = I'^2_{nL}(t_2) + V^2_{nC}(t_2) \tag{16}$$

To determine the transition from mode $M2$ to $M5$, the expression for i_{nLs} given in eqn. 7 is used to determine the conduction angle β of diode D_2. From the trajectory for the discontinuous conduction mode shown in Fig. 4, we obtain

$$\alpha + \beta + \gamma\frac{\omega_0}{\omega_p} = \pi \tag{17}$$

where γ is the total conduction angle for one cycle.

4 Performance characteristics

4.1 Control curves

In the series resonant convertor, power is coupled from the resonant tank to the output load through the resonant current. In the LLC-type SRC, the normalised output current can be obtained from the resonant current, i_{nLs}, by using the following relation:

$$I_{no} = \frac{2}{T_s}\int_{t_0}^{t_0+T_s/2} |i_{nLs}(t)|\, dt \tag{18}$$

The normalised output voltage, which is equal to the convertor gain, is given by

$$V_{n0} = M = \frac{I_{n0}}{Q} \tag{19}$$

where Q is the load quality factor given by Z_0/R_0, where R_0 is the load resistor. From eqns. 18 and 19, Q is evaluated from the following relations:

$$Q = \frac{\omega_0}{M\pi}\int_{t_0}^{t_0+T_s/2} |i_{nLs}(t)|\, dt \tag{20}$$

In terms of current responses in different circuit modes, eqn. 20 can be written as in eqns. 21 and 22 to correspond to the continuous and discontinuous conduction modes, respectively:

$$Q = \begin{cases} \dfrac{\omega_0}{M\pi}\displaystyle\sum_{i=0}^{2}\int_{t_i}^{t_i+1} |i_{nLs}(t)|\, dt \\ \dfrac{\omega_0}{M\pi}\displaystyle\sum_{i=0}^{1}\int_{t_i}^{t_i+1} |i_{nLs}(t)|\, dt \end{cases} \tag{21}$$

Substituting eqn. 6 into eqn. 21, we obtain two expressions which have the following functional form

$$Q = Q(\alpha, \beta, i_{nLs}(t_0), v_{nC}(t_0), M) \tag{22}$$

From the geometric properties of the state–plane diagrams given in Figs. 3 and 4 and from eqns. 7, 8 and 17, it can be shown that following functions can be obtained

$$\beta = \beta(\alpha, i_{nLs}(t_0), i_{nL}(t_0), v_{nC}(t_0), M) \tag{23}$$

$$\gamma = \gamma(\alpha, \beta) \tag{24}$$

By relating the switching points at the starting and ending circuit mode in the state–plane diagram, and using the circuit equations, we obtain the three steady-state conditions which satisfy the following functional forms

$$i_{nL}(t_0) = f_1(\alpha, \beta, \gamma, i_{nLs}(t_0), i_{nL}(t_0), v_{nC}(t_0), M) \tag{25}$$

$$v_{nC}(t_0) = f_2(\alpha, \beta, \gamma, i_{nLs}(t_0), i_{nL}(t_0), v_{nC}(t_0), M) \tag{26}$$

$$i_{nLs}(t_0) = f_3(\alpha, \beta, \gamma, i_{nLs}(t_0), i_{nL}(t_0), v_{nC}(t_0), M) \tag{27}$$

In eqns. 22–27, we have eight unknowns in six equations. Specifying any two of these unknown parameters, we can solve these equations numerically.

The control characteristics of the PWM-controlled LLC-type SRC have been obtained for diffeent values of Q and are given in Figs. 6, 7 and 8 for $k = 0.25$, 0.5, and 1, respectively. It is found, from these characteristics, that the convertor becomes load independent when it operates in the continuous conduction mode. For a given convertor gain and load range, it can be seen, from this figure, that the control characteristics of the LLC-type SRC are much improved over those of the conventional SRC. Note that the control duty ratio D shown on these control characteristic curves is defined as the ratio of the on-time of the switching transistor to one-half of the switching period. For example, for $k = 0.25$, $M = 0.4$ and a load range from $6/\pi$ to infinity, the control duty ratio is from 0.375 to 0.5. In the conventional SRC, the range of duty ratio is 0.21 to 0.5. There is about a 50% reduction

in control effort in regulating the gain over the same load range.

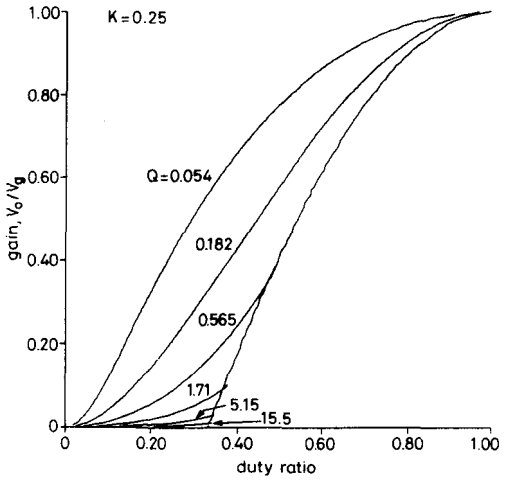

Fig. 6 *Control characteristic curves for K = 0.25*

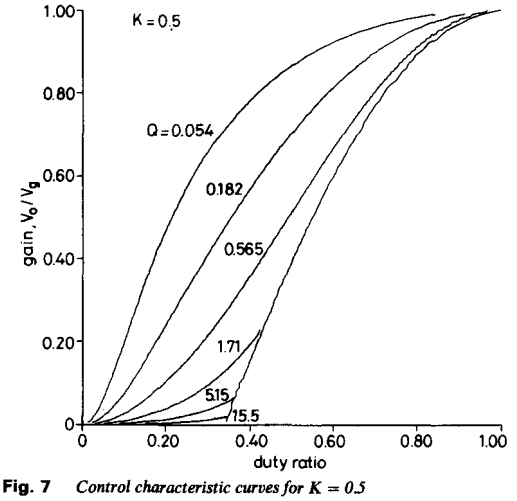

Fig. 7 *Control characteristic curves for K = 0.5*

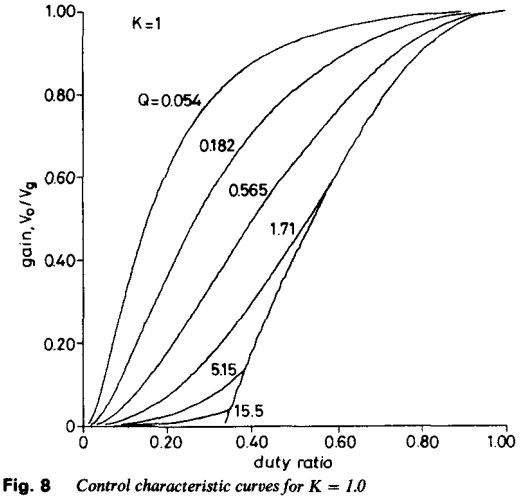

Fig. 8 *Control characteristic curves for K = 1.0*

5 Component stresses

In this Section, we present the component stress curves for the *LLC*-type SRC and then compare them with the component stress curves for the conventional SRC with PWM control, reported previously in Reference 3. The main interest here is to compute the peak current, $I_{nLs\,max}$, through the switching transistors and the peak voltage across the resonant capacitor, $V_{nC\,max}$. For a given convertor gain M, the peak current $I_{nLs\,max}$ and the peak capacitor voltage $V_{nC\,max}$ can be numerically calculated from eqn. 6, and eqns. 23–28, and are shown in Figs. 9 and 10 for $k = 0.25$. For $k = 0.5$, the component stress curves are given in Figs. 11 and 12. For comparison, the stress curves for the conventional SRC are shown in Figs. 13 and 14. From these Figures, it can be seen that $I_{nLs\,max}$ in the *LLC*-type SRC is less than that of the conventional SRC for a given gain M and duty ratio D. From Figs. 10 and 14, for a given M, we find that the voltage stress curves for both convertors are almost identical.

Under different load conditions, the curves for $I_{nLs\,max}$ and $V_{nC\,max}$ as a function of the duty ratio, d, are also plotted in Figs. 9–13. Again, $I_{nLs\,max}$ in the *LLC*-type SRC is less than that of the conventional SRC. However, from Figs. 10 and 14, $V_{nCs\,max}$ in the *LLC*-type SRC is greater than that of the conventional SRC. From Figs. 9–14, we find that, for a given gain M and load condition Q, the operating duty ratio d in the *LLC*-type SRC is

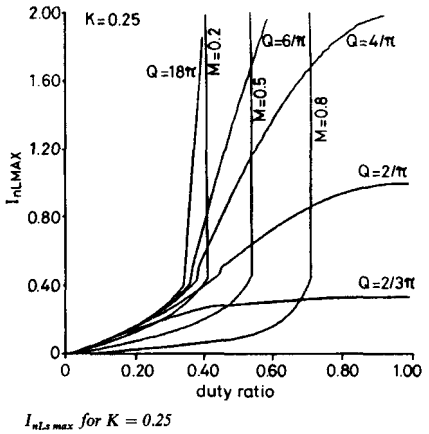

Fig. 9 *$I_{nLs\,max}$ for K = 0.25*

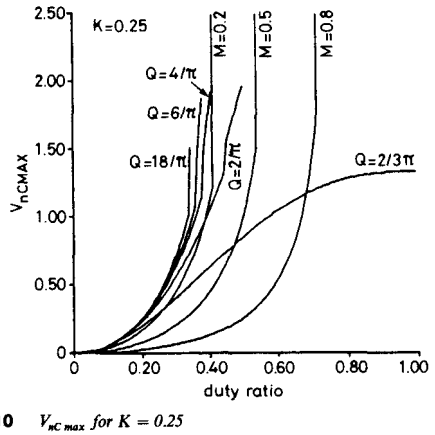

Fig. 10 *$V_{nC\,max}$ for K = 0.25*

higher. The slopes in the peak stress curves are discontinuous, as shown in Figs. 9–14. This is due to the transition periods between the discontinuous and the continuous conduction modes.

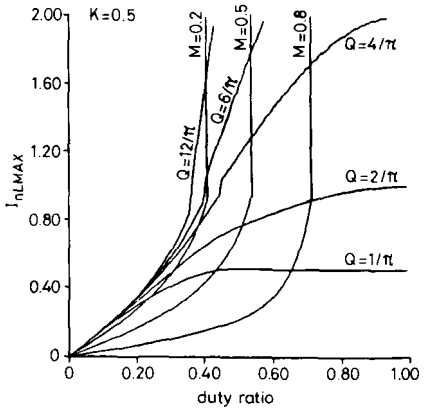

Fig. 11 $I_{nLs\,max}$ *for* $K = 0.5$

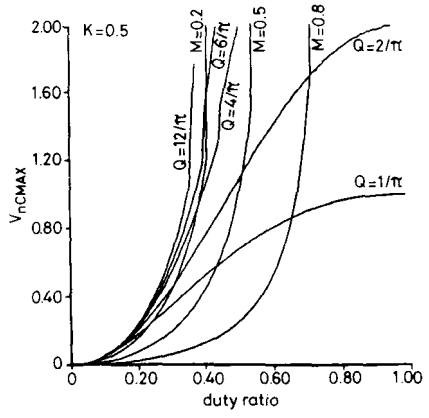

Fig. 12 $V_{nCp\,max}$ *for* $K = 0.5$

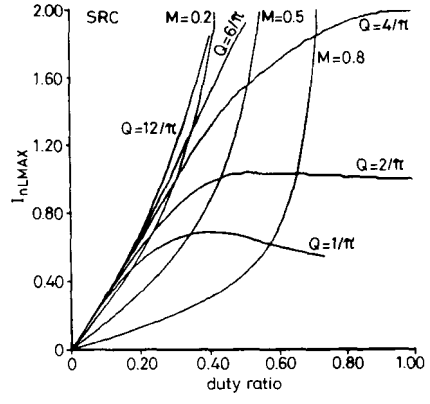

Fig. 13 $I_{nLs\,max}$ *for* $K = 1$ *(conventional SRC)*

6 Simulation and experimental results

To verify our theoretical work, a simulation program for the *LLC*-type SRC, operating at a duty ratio of 0.5 and under different load conditions, was performed. The theoretical and simulated results are given in Table 1. From these results it can be seen that the theoretical calculation agrees with the simulation results. Simulated waveforms and their trajectory are depicted in Figs. 15*a* and *b*.

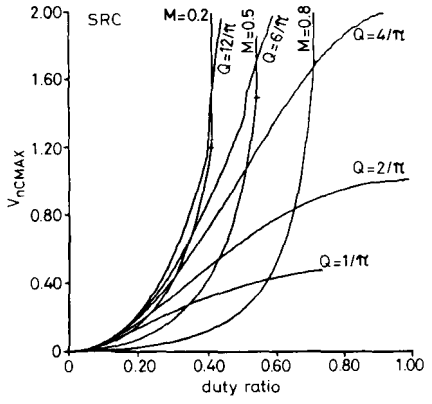

Fig. 14 $V_{nCp\,max}$ *for* $K = 1$ *(conventional SRC)*

To make a further comparison of the range of duty ratio needed for output regulation between the *LLC*-type SRC and the conventional SRC, a PSPICE program was used to simulate both convertors, with two different values of Q. For the *LLC*-type SRC with $k = 0.25$, the simulation waveforms for $Q = 2/\pi$ and $20/\pi$ are shown in Figs. 16 and 17, respectively. The corresponding waveforms for the conventional SRC ($k = 1$) are given in Figs. 18 and 19. The relevant results are given in Table 2. It can be seen from this Table that the range of duty ratio needed in the *LLC*-type SRC is only 58% of that in the conventional SRC.

A prototype *LLC*-type SRC with half-bridge topology was built and tested to verify the mode of operation discussed in this paper. With $R_0 = 27\ \Omega$, the waveforms and state–plane trajectories are shown in Figs. 20–22. Fig. 20 shows that the convertor is operated in the continuous conduction mode, which is also verified from the i_{Ls}

Table 1: Theoretical and simulated results for an *LLC*-type SRC at 0.5 duty ratio

Theoretical values	Simulated results
$Q = \frac{2}{\pi}$	$Q = \frac{2}{\pi}$
$M = 0.49$	$M = 0.481$
$V_{nC\,max} = 1.086$	$V_{nC\,max} = 1.08$
$I_{nL\,max} = 0.755$	$I_{nL\,max} = 0.761$
$Q = \frac{4}{\pi}$	$Q = \frac{4}{\pi}$
$M = 0.415$	$M = 0.414$
$V_{nC\,max} = 1.87$	$V_{nC\,max} = 1.95$
$I_{nL\,max} = 1.13$	$I_{nL\,max} = 1.17$
$Q = \frac{6}{\pi}$	$Q = \frac{6}{\pi}$
$M = 0.415$	$M = 0.414$
$V_{nC\,max} = 2.79$	$V_{nC\,max} = 2.73$
$I_{nL\,max} = 1.53$	$I_{nL\,max} = 1.51$

waveforms shown in Fig. 22. For light load, $R_0 = 500\ \Omega$, the convertor operates in the discontinuous conduction mode and the corresponding waveforms and trajectory are shown in Figs. 23–25. Based on the theoretical results, it is expected that i_{Ls} remains at zero once it reaches zero; however, the experimental measurements show a non-zero current before turning on the switch. This discrepancy is due to the parasitic capacitances of the power semiconductor switches. In general, we found that the component stresses in the practical circuit never exceeded the theoretical values, because of distributed losses in all components.

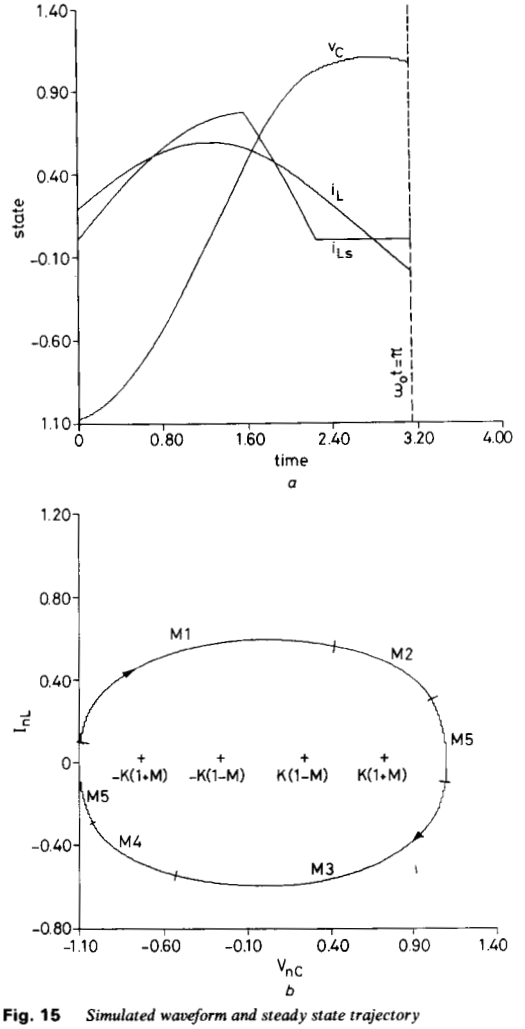

Fig. 15 *Simulated waveform and steady state trajectory*
a v_C, i_{Ls}, and i_L is steady state
b Simulated trajectory
($K = 0.5$ $M = 0.49$ $Q = 0.67$ $d = 0.5$)

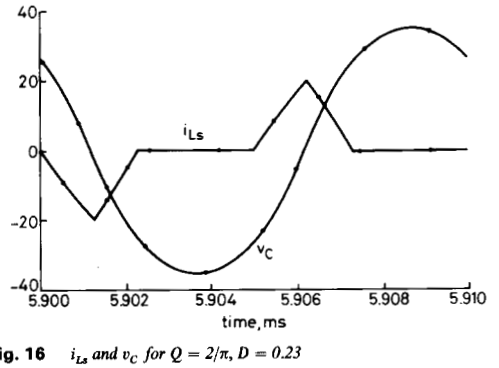

Fig. 16 i_{Ls} *and* v_C *for* $Q = 2/\pi$, $D = 0.23$
($K = 0.25$ $V_0 = 7.6$ V $R_L = 80.6\ \Omega$ $Z_0 = 51.3\ \Omega$)

7 Conclusions

It has been shown that, in the continuous conduction mode of operation, the control characteristic curves of

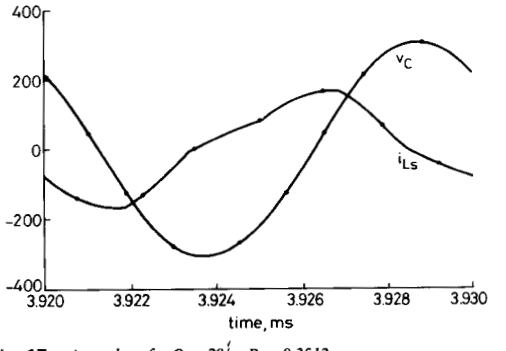

Fig. 17 i_{Ls} *and* v_C *for* $Q = 20/\pi$, $D = 0.3512$
($K = 0.25$ $V_0 = 7.6$ V $R_L = 8.06\ \Omega$ $Z_0 = 51.34\ \Omega$)

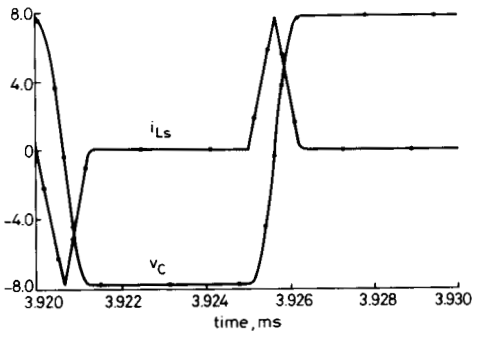

Fig. 18 i_{Ls} *and* v_C *for* $Q = 2/\pi$, $D = 0.114$
($K = 1$ $V_0 = 7.65$ V $R_L = 80.6\ \Omega$ $Z_0 = 51.3\ \Omega$)

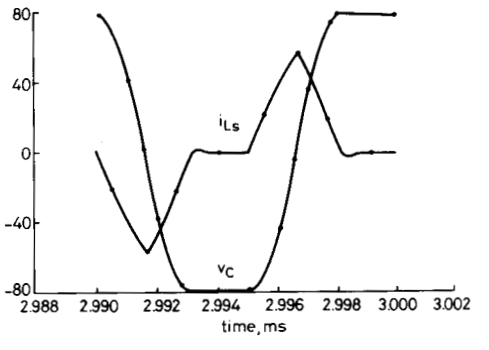

Fig. 19 i_{Ls} *and* v_C *for* $Q = 20/\pi$, $d = 0.321$
($K = 1$ $V_0 = 7.86$ V $R_L = 8.06\ \Omega$ $Z_p = 51.3\ \Omega$)

Table 2: Simulation results for *LLC*-type SRC versus conventional SRC

LLC-type ($K = 1/4$)	SRC ($K = 1$)
$Q = \frac{2}{\pi}$	$Q = \frac{2}{\pi}$
$M = 0.076$	$M = 0.0765$
$D = 0.23$	$D = 0.114$
$V_{nC\,max} = 0.352$	$V_{nC\,max} = 0.077$
$I_{nLs\,max} = 0.2054$	$I_{nLs\,max} = 0.3943$
$Q = \frac{20}{\pi}$	$Q = \frac{20}{\pi}$
$M = 0.076$	$M = 0.0786$
$D = 0.3512$	$D = 0.321$
$V_{nC\,max} = 3.093$	$V_{nC\,max} = 0.789$
$I_{nLs\,max} = 0.8625$	$I_{nL\,max} = 1.478$

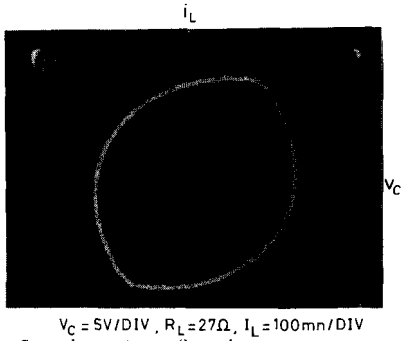

Fig. 20 *State-plane trajectory (heavy load)*
$R_L = 27\,\Omega$ $i_L = 100$ mA/div $v_C = 5$ V/div

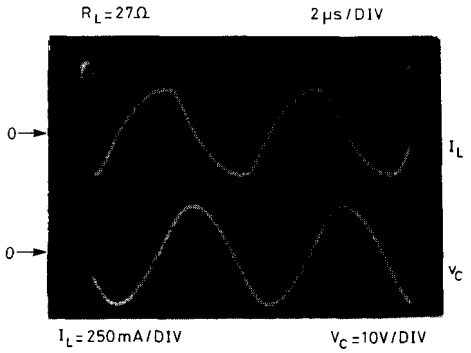

Fig. 21 *I_L and v_C waveforms (heavy load)*
$R_L = 27\,\Omega$ time = 2 μs/div $I_L = 250$ mA/div $v_C = 10$ V/div

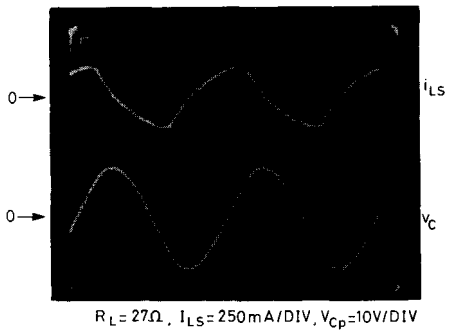

Fig. 22 *i_{Ls} and v_C waveforms (heady load)*
$R_L = 27\,\Omega$ time = 2 μs/div $i_{Ls} = 250$ mA/div $v_C = 10$ V/div

the *LLC*-type SRC cover a wider load range for a given voltage gain than the conventional SRC using PWM control. In the discontinuous conduction mode of operation, for a given voltage gain and under the same load range, the control characteristic curves of the *LLC*-type SRC possess a narrower range of duty ratio than the conventional SRC.

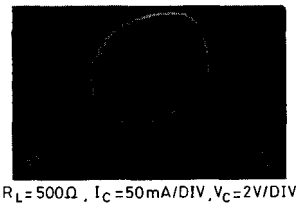

Fig. 23 *State-plane trajectory (light load)*
$R_L = 500\,\Omega$ $i_L = 50$ mA/div $v_C = 2$ V/div

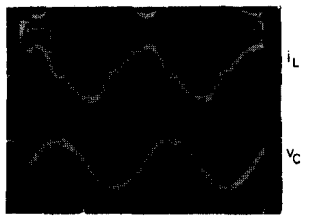

Fig. 24 *i_L and v_C waveforms (light load)*
$R_L = 500\,\Omega$ time 2 μs/div $i_L = 100$ mA/div $v_C = 5$ V/div

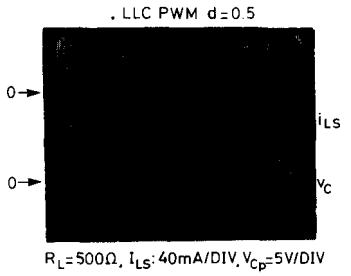

Fig. 25 *i_{Ls} and v_C waveforms (light load)*
$R_L = 500\,\Omega$ time 2 μs/div $i_{Ls} = 40$ mA/div $v_C = 5$ V/div

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